

Parameter Optimization of Linear Quadratic Controller Based on Genetic Algorithm

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Abstract: The selection of weighting matrix in design of the linear quadratic optimal controller is an important topic in the control theory. In this paper, an approach based on genetic algorithm is presented for selecting the weighting matrix for the optimal controller. Genetic algorithm is adaptive heuristic search algorithm premised on the evolutionary ideas of natural selection and genetic. In this algorithm, the fitness function is used to evaluate individuals and reproductive success varies with fitness. In the design of the linear quadratic optimal controller, the fitness function has relation to the anticipated step response of the system. Not only can the controller designed by this approach meet the demand of the performance indexes of linear quadratic controller, but also satisfy the anticipated step response of close-loop system. The method possesses a higher calculating efficiency and provides technical support for the optimal controller in engineering application. The simulation of a three-order single-input single-output (SISO) system has demonstrated the feasibility and validity of the approach.

Key words: genetic algorithm; weighting matrix; linear quadratic controller; parameter optimization

Introduction

The design of the optimal controller is an important subject in the optimization theory. In the design of a quadratic optimal controller, two primary problems have not been solved. One is how to get the state weighing matrix and control weighing matrix in the light of the time domain index, and the other is how to solve Riccati equation to get the optimal results. The design of the state weighing matrix is the key problem in the design of a quadratic optimal controller. The traditional methods are experimentation, optimization, method of approximation, and optimal pole collocation method. Although these methods get some achievements, they have limitation in a certain extent. In this paper, genetic algorithm is adopted to solve the prob-

lem.

Genetic algorithm is an evolution method based on natural selection theory and biological genetics. The algorithm simulates reproduction, intercross, and aberrance in the processing of natural selection and natural inheritance. According to the natural rule “the survival of the fittest”^[1], it uses genetic operators to select intercross and mutation until the optimal results are gained. In the last twenty years, genetic algorithm has been applied in all kinds of disciplines, such as machine learning, image processing, neural networks, pattern reorganization, and social science. Genetic algorithm provides a new way for the design of the optimal controller.

1 Linear Quadratic Optimal Controller

The state equation for the linear time-invariant system is expressed as^[2]

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$$\dot{x}(t) = Ax(t) + Bu(t) \quad (1)$$

where $x(t)$ is n -dimensional state vector, $u(t)$ is m -dimensional control vector ($m \leq n$), A is $n \times n$ -dimensional time-invariant array, and B is $n \times m$ -dimensional time-invariant array. Suppose that the control vector $u(t)$ is free.

The state feedback regulator $u^*(t)$ is to be designed by minimizing the quadratic performance index given by

$$J = \frac{1}{2} x^T(t) F x(t) + \frac{1}{2} \int_{t_0}^{t_f} [x^T(t) Q x(t) + u^T(t) R u(t)] dt \quad (2)$$

where F is $n \times n$ -dimensional positive semi-definite constant weighing matrix. Q is $n \times n$ -dimensional positive semi-definite time-invariant state weighing matrix. R is $m \times m$ -dimensional positive semi-definite time-invariant control weighing matrix. The start time t_0 and the end time t_f are fixed.

The control gain matrix K is given by

$$u^*(t) = -R^{-1} B^T P x(t) = -K x(t) \quad (3)$$

$$PA + A^T P - P B R^{-1} B^T P + Q = 0 \quad (4)$$

where P is $n \times n$ -dimensional unique positive definite solution to the matrix Riccati equation as we discussed above.

In this problem, the first item on the right of the Eq. (2) is terminal cost. It stands for the similarity between the system end state and the anticipated state when the end time t_f is coming. The integral item on the right of the Eq. (2) is general index. Where the first item $\frac{1}{2} x^T(t) Q x(t)$ is a restriction for the state $x(t)$ and it can show the system general error between the given state and the practical state in the control processing. The second item stands for a restriction index in dynamic process, as well as a restriction index for the energy in the control processing. In fact, two integral items are restricted each other. In order to reduce the error index of the control state, we have to expend much more control energy. On the contrary, for saving control energy, we must give up the demand of the control performance indexes. In the project, how to select elements in the weighing matrix F , Q , and R according to the demand of control system is an important and difficult job^[3]. In the following, genetic algorithm is adopted to design the weighing matrix for the optimal controller.

2 Choice of Weighting Matrix in Design of the Linear Quadratic Optimal Controller

Because the weighing matrix Q and R are relative, in general, we can fix matrix R and select the optimal elements in the matrix Q . When the system has a single input, R is a scalar quantity. So we can select $R = [1]$.

Suppose that Q meets the demands of the system. For Q given randomly, the optimal feedback control vector $u^*(t) = -Kx$ can make the close-loop control system $\dot{x}(t) = (A - BK)x$ stable gradually in a large-scale. Suppose the step response of close-loop system designed by the given matrix Q is y_1 and the anticipated step response of close-loop system is y_2 . We define

$$f = \|y_1 - y_2\|_1$$

In this way, we can select the weighing matrix Q according to the formula above.

The processing for selecting the weighing matrix is described as follows. At first, we need to select the initial value by our experience and code them. The weighing matrix Q is gained by the code. Then, we can simulate by the weighing matrix and the simulation results are validated by the fitness function. The fitness function is the formula as we defined above. It can reflect the relations between the system step response, and then the anticipated response. The individual with lower fitness is given up and the individual with higher fitness is selected and become the new individual in next generation. After performing crossover and mutation, the new generation is produced. Repeat steps as we presented above. The fitness of the new generation is always higher than the last generation. The process is repeated again and again until the optimal result is found. In the following, we will explain the whole process by an example of a real system.

3 Example of a Real System

The linear system^[4] to be designed is

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx + Du \end{cases}$$

where

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -2 & -3 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, C = [1 \quad 0 \quad 0], D = 0$$

The state feedback regulator u is to be designed by minimizing the quadratic performance index given by

$$J = \int_0^\infty (\mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{u}^T \mathbf{R} \mathbf{u}) dt$$

where $\mathbf{R} = [0.01]$, $\mathbf{Q} = \begin{bmatrix} q_{11} & 0 & 0 \\ 0 & q_{22} & 0 \\ 0 & 0 & q_{33} \end{bmatrix}$.

We need to select the appropriate elements in the matrix $\mathbf{Q}$ in order to make the system step response close to the anticipated step response.

The process is organized as follows:

The first step is building up the mathematical model with respect to the problem. The mathematical model is directly given in this scenario.

The second step is coding. The floating-point coding method is a good choice because it has advantage in efficiency, precision, and exaction. In this method, every element in the individual is presented by a floating-point number in a certain range, and the length of the individual coding is equal to the number of the decision-making variable. The real decision-making variable is adopted in the coding method, so the floating-point coding method is also called the real value coding. We define $\mathbf{Q} = [q_{11}, q_{22}, q_{33}]^T$. The ranges of elements are $q_{11} \in [100, 200]$, $q_{22} \in [1, 10]$, $q_{33} \in [0.01, 0.1]$.

The third step is producing the initial group. This step is including two contents: one is the size of the group; the other is the selection of the initial group. The size of the group can affect the final results and the efficiency of genetic algorithm. If the size is too small, the performance of genetic algorithm is not good. On the contrary, the great group can improve the performance, but it means a mass of calculations. Generally, the size of the group is from 10 to 160. In this example, the size of the group is 50. The initial group is produced randomly.

The fourth step is to select the fitness function. Suppose that the open-loop transfer function of the close-loop system is $\mathbf{G}(s)$.

$$\mathbf{G}(s) = \mathbf{K}(s\mathbf{I} - \mathbf{A})^{-1} \mathbf{B} \quad (5)$$

where the matrix $\mathbf{K}$ is the state feedback gain matrix.

Suppose the step response of close-loop system de-

signed by the given matrix $\mathbf{Q}(t)$ is $\mathbf{y}_1$ and the anticipated step response of close-loop system is $\mathbf{y}_2$.

$$\mathbf{Y}_1(s) = \frac{1}{1 + \mathbf{G}(s)} \times \frac{1}{s} \quad (6)$$

$$\mathbf{Y}_2(s) = \frac{\omega^2 s_0}{(s + s_0)(s^2 + 2\xi\omega s + \omega^2)} \times \frac{1}{s} \quad (7)$$

where $s_0 = 100$, $\xi = 0.707$, $\omega = 2$ rad/s.

We can get the step response $\mathbf{y}_1$ by transforming the form of $\mathbf{Y}_1(s)$ using Laplace inverse theory. Also, $\mathbf{y}_2$ is gained by transforming the form of $\mathbf{Y}_2(s)$ using Laplace inverse theory. The fitness function is given as following.

$$F = \frac{1}{1 + \|\mathbf{y}_1 - \mathbf{y}_2\|_1} \quad (8)$$

The fitness function is given in this way in order to avoid converging quickly.

The fifth step is selecting operation. The proportion selection method adopted by the example is a random sampling way. Suppose that the size of the group is M and the fitness of an individual is F_i . The selection probability is

$$p = F_i / \sum_{i=1}^M F_i (i = 1, 2, 3, \dots, M) \quad (9)$$

The individual with lower fitness is given up and the individual with higher fitness is selected, and then become the new individual in next generation.

The sixth step is crossover. The arithmetic crossover is a good choice for this case. The new individual is produced by linear operations of the old individuals. The two former individuals are q_1^t , q_2^t and the new individuals are q_1^{t+1} , q_2^{t+1} , respectively. The selected operation is

$$\begin{cases} q_1^{t+1} = \alpha q_2^t + (1 - \alpha) q_1^t \\ q_2^{t+1} = \alpha q_1^t + (1 - \alpha) q_2^t \end{cases} \quad (10)$$

where α is constant, $\alpha = 0.4$.

The last step is aberrance operation. The aberrance operation can help genetic algorithm search results randomly and maintain the diversity of the group. We can select some individuals randomly from the group and calculate the opposite values.

After aberrance operation, we can get a new group. Then, the new group repeats the process as we discussed above. The process is repeated again and again until the optimal result is found.

After a series of operations, the optimal weighing matrix for the linear control system is

$$\mathbf{Q} = \begin{bmatrix} 193.18 & 0 & 0 \\ 0 & 1.0148 & 0 \\ 0 & 0 & 0.0551 \end{bmatrix}.$$

The corresponding feedback gain matrix is $\mathbf{K} = [138.99 \ 54.529 \ 8.116]$.

As shown in Fig. 1, the broken line stands for the anticipated response and the real line stands for the system step response. The two lines are not superposable, but they are resemble, which shows that the system step response is close to the anticipated response of the loop-close system. The dynamic performance between the anticipated response and the system step response is given in Table 1.

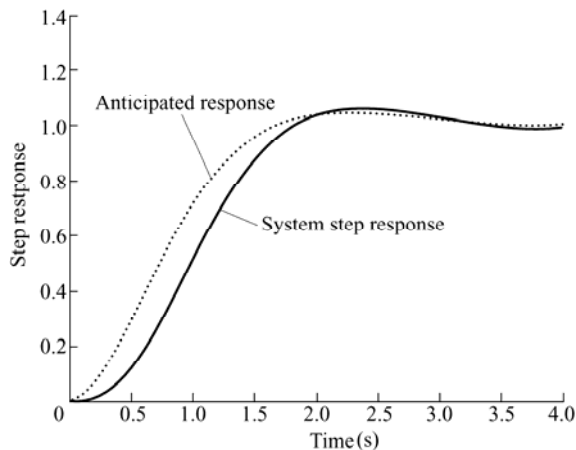


Fig. 1 System step response curve and the anticipated response curve.

Table 1 Dynamic performance between the anticipated response and the system step response

Index	Rise time(s)	Adjustment time(s)	Over-shoot	Extreme point
Anticipated response	1.6657	2.4752	0.0433	-100.00 -1.414+1.414i -1.414-1.414i -5.4173
Step response	1.0934	2.6163	0.0603	-2.8494+4.1878i -2.8494-4.1878i

4 Conclusions

From the Fig. 1 and the Table 1 given above, we can make a conclusion that the controller designed by this

approach can not only meet the demand of the performance index of linear quadratic controller, but also can satisfy the anticipated step response of close-loop system. The approach presented in this paper has certain instructive significance for engineering project.

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